

Labwork 6.2 - Letter Count

General Information and Submission Rules:

- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork6.c**. In addition to this, you need to create a **command line history** file with this command:
history > history_fullname_labwork6.txt
and submit this also.
- Incorrect naming or file format will result in **a loss of 5 points** from your total grade.
- Only use concepts covered in the labs so far.
- You can put these given values into your main directly and assume the alphabet is an English alphabet: (When you copy and paste it to your code, check the correctness of it);

```
char sentence[91] = "silent shadows waltz with the moons gentle glow painting  
poetry upon the canvas of the night";  
int count[26] = {0}; // It will keep the occurrences of each letter char and  
initially all of them is zero  
alphabet[26] = { 'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm',  
'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z' };
```

Question 1 - Display Message (25p)

- Create a new function (**welcome**) that displays a welcome message and a list of all available operations to the user.
- This function will expect a decision from a user so that it will execute related functions based on this choice until the user exits.
- Make sure to print each line separately.

```
Welcome to the Letter Count Machine! By using this machine you can do:  
1. Increment Letter  
2. Count Letter  
3. Find Most Frequent Letter  
4. Display Prime Occurrences (Bonus)  
5. Exit  
Make your choice:
```

Question 2 - Increment Letter (25p)

- Create a new function (**increment_letter**) that takes the array of letters, count array that keeps the occurrences of these letters, the letter that needs to be incremented, and how many times you need to increment that letter as parameters.

Input: letter_array, count_array, letter, increment_value

Output: print count_array

- In the end this function will increment the occurrence value of that letter.

```
>> Enter the letter: a
>> Enter increment value: 1
a - 1
```

Question 3 - Count Letter (25p)

- Create a new function (**count_letter**) that takes the array of letters, given string and count array that keeps the occurrences of these letters as parameters.
- In the end, this function will find out the occurrences of each letter in a given string.
- Only print letters that were found at least once.

```
Letter Occurrence Count:
a: 5
c: 1
d: 1
e: 7
f: 1
g: 4
h: 6
i: 5
l: 4
m: 1
n: 8
o: 7
p: 3
r: 1
s: 5
t: 10
u: 1
v: 1
w: 4
y: 1
z: 1
```

Question 4 - Find Most Frequent Letter (25p)

- Create a new function (**most_frequent**) that takes the array of letters, and count array that keeps the occurrences of these letters as parameters.
- In the end, this function will display the most frequent letter that occurred in a given string.

```
t - 10 times
```

Bonus - Display Prime Occurrences

- Create a new function (**prime_occur**) that takes the array of letters, and count array that keeps the occurrences of these letters as parameters.
- In the end, this function will display letters that appeared a prime number of times.

```
Prime Occurrences:  
a: 5  
e: 7  
i: 5  
o: 7  
p: 3  
s: 5
```